

20260923 Quiz

for (MATH200) Differential Equations

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Problem 2.4.1. Determine, without solving the problem, an interval in which the solution of the initial value problem

$$(t - 5)y' + (\ln t)y = 2t, \quad y(1) = 2,$$

is certain to exist.

Solution. Dividing by $t - 5$ gives the standard form

$$y' + \frac{\ln t}{t - 5}y = \frac{2t}{t - 5}.$$

The coefficients are continuous where $t > 0$ and $t \neq 5$. Thus the largest open interval containing the initial point $t = 1$ on which both coefficients are continuous is

$$(0, 5).$$

By Theorem 2.4.1, the solution is certain to exist throughout this interval.

□

Problem 2.4.2. Determine, without solving the problem, an interval in which the solution of the initial value problem

$$y' + (\tan t)y = \sin t, \quad y(2\pi) = 0,$$

is certain to exist.

Solution. Here

$$p(t) = \tan t, \quad g(t) = \sin t.$$

The function g is continuous on $\mathbb{R}$, while p is discontinuous at

$$t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Since

$$\frac{3\pi}{2} < 2\pi < \frac{5\pi}{2},$$

the largest open interval containing the initial point on which both coefficients are continuous is

$$\left(\frac{3\pi}{2}, \frac{5\pi}{2}\right).$$

By Theorem 2.4.1, the solution is certain to exist throughout this interval.

□

Problem 2.4.5. State where in the ty -plane the hypotheses of Theorem 2.4.2 are satisfied for

$$y' = \sqrt{4 - t^2 - y^2}.$$

Solution. Let

$$f(t, y) = \sqrt{4 - t^2 - y^2}.$$

Then

$$f_y(t, y) = -\frac{y}{\sqrt{4 - t^2 - y^2}}.$$

The function f is real-valued and continuous where $t^2 + y^2 \leq 4$, while f_y is continuous only where

$$t^2 + y^2 < 4.$$

Hence both f and f_y are continuous in a rectangle about a point precisely when the point lies in the open disk

$$\boxed{t^2 + y^2 < 4}.$$

Thus the hypotheses of Theorem 2.4.2 are satisfied exactly in this region.

□

Problem 2.6.1. Determine whether

$$(4x + 3) + (6y - 1)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 4x + 3, \quad N(x, y) = 6y - 1.$$

Then

$$M_y = 0 = N_x,$$

so the equation is exact. Let $\psi_x = M$ and $\psi_y = N$. Integrating $\psi_x = M$ with respect to x gives

$$\psi(x, y) = 2x^2 + 3x + h(y).$$

Therefore

$$\psi_y = h'(y) = 6y - 1,$$

so

$$h(y) = 3y^2 - y.$$

Hence the solution is given implicitly by

$$\boxed{2x^2 + 3x + 3y^2 - y = c}.$$

□

Problem 2.6.2. Determine whether

$$(3x - y) + (x - 3y)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 3x - y, \quad N(x, y) = x - 3y.$$

Then

$$M_y = -1, \quad N_x = 1.$$

Since $M_y \neq N_x$, the equation is

not exact

.

□

Problem 2.6.3. Determine whether

$$(6x^2 - 2xy + 4) + (6y^2 - x^2 + 2)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 6x^2 - 2xy + 4, \quad N(x, y) = 6y^2 - x^2 + 2.$$

Then

$$M_y = -2x = N_x,$$

so the equation is exact. Let $\psi_x = M$ and $\psi_y = N$. Integrating $\psi_x = M$ with respect to x gives

$$\psi(x, y) = 2x^3 - x^2y + 4x + h(y).$$

Therefore

$$\psi_y = -x^2 + h'(y) = 6y^2 - x^2 + 2,$$

which gives

$$h'(y) = 6y^2 + 2, \quad h(y) = 2y^3 + 2y.$$

Hence the solution is given implicitly by

$$\boxed{2x^3 - x^2y + 4x + 2y^3 + 2y = c}.$$

□